Doubly Robust Estimators for Survival Outcomes: Point Exposures and Time-fixed Confounders

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7.1.2 Inverse Probability Treatment Weighting (IPTW)

1 Introduction

This document provides a summary of doubly robust estimators for survival outcomes. We focus on the application of these methods to observational / real-world data i.e. in non-randomised studies. Therefore, the goal is to minimise bias in our target estimand when estimating a causal effect. In the absence of randomisation, we rely on the correct specification of the outcome model and that all confounders have been appropriately adjusted for. The marginal causal effect is then obtained based on the assumption that this outcome regression model is correctly specified. Alternatively, we can model the exposure itself to ensure confounders are balanced between the exposed and unexposed groups. This relies on the correct specification of the exposure model. However, a “doubly-robust” estimator provides a consistent estimate of the causal effect if at least one of either the outcome or exposure model is correctly specified and if both are correctly specified then it provides an unbiased estimate. In other words, this provides two shots at the target. We can achieve both consistency and efficient estimate, if both models are specified correctly. In a well-designed trial, due to randomisation, the exposed and unexposed arms are balanced so the estimates are consistent, and doubly robust estimation provides improvements in efficiency. This is not the case for non-randomised studies, and thus, we intend to compare various doubly-robust methods with the goal of evaluating their performance and robustness to model misspecification in at least one of the outcome or exposure model.

2 Literature Review

In recent years, doubly robust estimation has become a hot-topic. It has mainly gained traction due to its attractive feature of obtaining maximally efficient estimates of the target estimand in randomised clinical trials after covariate adjustment. This enables reductions in standard errors, leading to lower sample sizes, or increased power. In the context of non-randomised studies, they provide the additional benefit of maximal efficiency through robustness to model misspecification for the estimation of marginal causal effects. In literature, there are several papers that make comparisons between doubly robust estimators for continuous and binary endpoints in the context of non-randomised studies (refs).

However, there have been few publications

We have identified commonly used methods in practice and have selected those that do not have a complex implementation and are accompanied by software for practical applicability:

- Inverse Probability Treatment Weights
- Regression Standardisation / Simple G-computation
- Doubly Robust Standardisation / IPTW-RA
- Augmented Inverse Probability Treatment Weights
- Targeted Maximum Likelihood Estimation

3 Causal Framework and Estimands

3.1 Notation

Let us define some notation and assumptions

- X: the vector of measured baseline covariates
- A: a binary indicator for the treatment received, with $A = 0$ denoting the control treatment (e.g., placebo) and $A = 1$ the active treatment.
- $h(t|A = a)$: the observed hazard function at time t among individuals who actually received treatment $A = a$.
- $h_a(t)$: the potential (counterfactual) hazard function at time t that would arise if every individual were assigned treatment $A = a$.

3.2 Causal estimands of interest

Various estimands can be used to summarize treatment effects in time-to-event analyses. In this work, we adopt a marginal causal estimand framework that focuses on population-level survival quantities defined under hypothetical treatment interventions. These estimands describe the expected survival experience of the target population if all individuals were assigned to a given treatment and are standardized over the observed distribution of baseline covariates rather than conditional on specific covariate values. The estimand of interest is the average treatment effect in the entire population (ATE).

The primary estimands considered are treatment-specific marginal survival curves and their corresponding cumulative incidence functions. These curves provide a covariate-adjusted description of the probability of remaining event-free, or experiencing the event, over time under each treatment strategy. Treatment effects are summarized by contrasts between these marginal curves, such as absolute differences or ratios in survival or risk at prespecified time points, as well as functionals that aggregate differences over time, including the restricted mean survival time. These summaries yield interpretable, population-averaged measures of treatment effect without reliance on proportional hazards or other structural assumptions.

For completeness, we note that a *marginal hazard ratio* can be defined as the hazard ratio for the exposure unconditional on other covariates, for example as estimated by a correctly specified Cox model containing only the exposure [1]. Although such hazard ratios can, in principle, be derived from marginal survival curves, their interpretation in a marginal causal framework is problematic. Accordingly, hazard ratios are not treated as primary estimands in the present analysis. The implications of this choice, particularly in the context of singly and doubly robust estimation, are discussed in Section 4.3.

4 Marginal Estimation Methods

4.1 Singly robust estimators

4.1.1 Inverse Probability Treatment Weighting (IPTW)

Inverse Probability of Treatment Weighting (IPTW) is a causal inference method used to address imbalances in baseline covariates between treatment groups when these covariates affect both treatment assignment and the outcome. The procedure involves estimating each individual's probability of receiving their observed treatment based on measured baseline characteristics, $e = \Pr(A = 1|X)$. Individuals are then weighted by the inverse of this probability, with the precise form of the weights depending on the target estimand [2]. These weights create a pseudo-population in which treatment assignment is independent of measured baseline covariates, thereby reducing confounding bias in the estimated treatment effect.

In time-to-event analyses, IPTW is commonly implemented in combination with a Cox proportional hazards model. In this setting, the Cox model is fitted in the weighted pseudo-population, typically with the treatment indicator as the sole covariate, and inference is obtained a robust variance estimator to account for the fact that the weights are estimated rather than known with certainty. Alternatively, nonparametric bootstrap procedures may be used to obtain standard errors and confidence intervals [3].

When estimating the average treatment effect in the entire population, individuals are weighted according to the inverse of their probability of receiving the treatment actually observed. Fitting a weighted Cox proportional hazards model under this specification yields an estimate of the marginal hazard ratio, defined as the hazard ratio for the exposure unconditional on baseline covariates. This estimand corresponds to a contrast between treatment-specific hazard functions in the IPTW pseudo-population.

While this approach is widely used in applied research, it is important to note that the hazard ratio constitutes a specific choice of estimand. As discussed in Section 3.2, and further elaborated in Section 4.3, hazard-based estimands raise important interpretational considerations in marginal causal frameworks.

As with all weighting approaches, IPTW estimates may become unstable when some individuals receive disproportionately large weights [2]. This typically occurs when estimated propensity scores are close to zero or one. Common strategies to mitigate the influence of extreme weights include weight stabilization, trimming, or truncation, and the use of data-adaptive models for propensity score estimation, such as random forests or gradient-boosted trees. Example code is provided in Section 8.1.1.

See ref: [4,5] Inverse Probability Weighting (IPW) – Advanced CI/ML Methods

- Issues with collapsibility
- Not compatible with PH models
- Also need IPCW
- Propose an alternative i.e. additive hazards models or pseudo observations

4.1.2 Regression Standardisation / G-computation

4.2 Doubly Robust Estimators

Estimator	Summary of method
DR-Standardisation	Calculate weights g -> weight data -> fit weighted outcome model Q -> standardise
AIPW	Prediction of outcome model Q_0 with weighted residuals using weights g
TMLE	Update outcome model Q_0 with a clever covariate from weights g -> standardise

4.2.1 Doubly Robust Standardisation

Doubly robust standardisation methods combine outcome regression and inverse probability weighting to estimate marginal causal estimands in time-to-event settings. These approaches are designed to estimate population-level survival quantities – such as treatment-specific survival curves, cumulative incidence functions, or functionals derived from these curves – under hypothetical treatment interventions, while offering protection against certain forms of model misspecification.

In survival analyses, right censoring and treatment-dependent selection over time complicate purely regression-based or purely weighting-based approaches. Outcome regression methods, such as g-computation or regression standardisation, rely on correct specification of the outcome model to extrapolate counterfactual survival distributions, whereas inverse probability weighting depends on correct specification of the treatment and censoring models and may suffer from instability when estimated weights are highly variable. Doubly robust standardisation addresses these limitations by combining outcome regression and weighting components within a single estimator [6].

Conceptually, doubly robust standardisation proceeds in two stages and is typically implemented using a discrete-time representation of the survival process in time-to-event settings. First, models are specified for the treatment assignment mechanism and, where relevant, the censoring process, yielding inverse probability weights that balance baseline covariates between treatment groups and adjust for informative censoring. Second, an outcome model for the discrete-time event process is fitted using these weights, and predictions from this model are averaged over the empirical covariate distribution to obtain marginal survival or cumulative incidence estimates. The resulting estimator is consistent for the target marginal estimand if either the outcome model or the treatment and censoring models are correctly specified, but not necessarily both.

Doubly robust standardisation is closely related to, but distinct from, targeted maximum likelihood estimation (TMLE). Both approaches combine outcome and treatment/censoring models and share the property of double robustness. However, whereas standardisation-based estimators typically update the outcome model through weighting and averaging, TMLE further targets the initial outcome estimate to solve the efficient influence function for the chosen estimand. In practice, both frameworks aim to estimate the same class of marginal survival estimands and differ mainly in their updating strategy and inferential machinery.

4.2.2 AIPTW

The AIPW estimator consists of two main components [7]. First, a propensity score model is fitted to estimate each individual's probability of receiving the treatment. When all confounding is captured by baseline covariates, this propensity score model is identical to the one used for binary outcomes, since treatment assignment occurs only once at baseline. Second, two outcome (or "response surface") regression models are fitted to estimate the counterfactual survival outcome under treatment and under control, adjusting for relevant baseline covariates. The AIPW estimator then combines ("augments") these outcome predictions with a correction term constructed from weighted residuals, where the weights are the inverse probabilities of the observed treatments (and, if applicable, of remaining uncensored when censoring is informative). This augmentation yields a doubly robust estimator that remains consistent if either the treatment model or the outcome model is correctly specified. This estimator is algebraically equivalent to the doubly robust sequential-regression (augmented G-computation) formulation of Bang and Robins [8], which embeds the same augmentation term within a regression framework.

For time-to-event (survival) outcomes, the outcome regression component must account for (i) the event process and (ii) the censoring mechanism (if censoring depends on covariates). This typically involves modelling the conditional hazard (or discrete-time hazard) for the event given baseline covariates and treatment; optionally, one also models the censoring process to compute inverse-probability-of-censoring weights (IPCW). Common choices for modelling the event process include:

- Pooled logistic regression (discrete-time hazard),
- Cox proportional hazards models,
- More flexible machine-learning models (e.g., survival forests, neural networks)

Under this setup, one can estimate treatment-specific survival curves (or restricted mean survival times), by combining (a) the hazard / survival predictions from the outcome model with (b) weights for censoring (if needed) and (c) weights for treatment (the propensity scores). The resulting estimator yields the marginal (population-level) survival curves under treatment vs. control, from which treatment effects (e.g., difference in survival probability at time t, or difference in restricted mean survival time) can be derived.

An advantage of the AIPW representation is that it makes the weighting mechanisms for confounding (and, if needed, censoring) explicit, while allowing the propensity-score model and the outcome models to be specified with different covariate sets. This flexibility enables consistent estimation as long as either component is correctly specified.

As with conventional IPTW, valid standard error can be obtained using an empirical sandwich (robust) variance estimator or through non-parametric bootstrapping [7].

4.2.3 TMLE

Targeted Maximum Likelihood Estimation (TMLE) is a semi-parametric causal inference method designed to estimate covariate-adjusted marginal outcome distributions under well-defined treatment strategies. TMLE is part of a broader class of causal methods developed to address confounding – particularly time-varying confounding – in longitudinal observational data, alongside approaches such as IPTW and g-computation [9].

In the context of time-to-event outcomes, TMLE targets the entire marginal survival function (or its complement, the cumulative incidence function) rather than a single regression coefficient. As a result, TMLE is particularly well suited for estimating population-level, treatment-specific survival curves and contrasts derived from these curves.

TMLE for time-to-event outcomes is typically implemented on a discrete-time scale, where follow-up is partitioned into a sequence of time intervals and the hazard of the event is modeled conditionally on remaining event-free and uncensored up to the start of each interval [10]. This formulation allows for flexible modeling of the outcome process while naturally accommodating right censoring and, if needed, time-varying covariates or treatments.

The TMLE procedure proceeds in two main steps. First, an initial outcome model is fitted to estimate the discrete-time hazard (or equivalently, the conditional survival probability) as a function of treatment and baseline or time-varying covariates. In parallel, models are specified for the treatment assignment mechanism and the censoring process, which together define the probability of remaining under observation and receiving the observed treatment history. These models may be estimated using parametric regression or more flexible, data-adaptive methods.

Second, the initial outcome model is updated through a targeting step, which incorporates information from the treatment and censoring models via a so-called clever covariate (corresponding to inverse-probability weights constructed from the treatment and censoring models). This targeted update ensures that the resulting estimator solves the estimating equation associated with the efficient influence function of the chosen estimand. As a result, TMLE yields doubly robust estimates: consistency is retained if either the outcome model or the treatment and censoring models are correctly specified. In addition, TMLE enables valid statistical inference using influence-function–based standard errors.

By construction, TMLE produces covariate-adjusted marginal survival curves or cumulative incidence functions that represent the expected survival experience of the study population under each treatment strategy, standardized over the empirical covariate distribution. Treatment effects can then be summarized using contrasts of these marginal curves, such as differences or ratios in survival probabilities at prespecified time points, or functionals of the curve such as the restricted mean survival time. Importantly, these estimands are marginal, clinically interpretable, and do not rely on proportional hazards or other strong modeling assumptions.

4.3 Interpretation of hazard ratios in marginal and doubly robust frameworks

Inverse probability weighting and doubly robust estimators such as augmented IPTW and TMLE are designed to estimate marginal causal estimands, including treatment-specific survival curves, cumulative incidence functions, and contrasts derived from these quantities. These estimands describe population-level survival experiences under hypothetical treatment interventions and admit a clear causal interpretation.

Although hazard functions and hazard ratios can be technically derived from marginal survival curves, their interpretation in a marginal causal framework is problematic. The hazard ratio compares instantaneous event rates among individuals who have survived up to a given time point and therefore depends on the composition of the time-specific risk set. Because survival itself is affected by treatment, the risk set evolves differently across treatment groups, inducing treatment-dependent selection over time. As discussed previously [11–13], this implies that hazard ratios reflect contrasts between dynamically selected subpopulations rather than a fixed population-level causal contrast, a phenomenon often referred to as collider (or survivor) bias.

These features imply that hazard ratios do not correspond to a simple or time-invariant marginal causal estimand tied directly to the survival distribution. Their magnitude and interpretation may vary over time, even when differences in marginal survival or cumulative incidence remain stable. Consequently, hazard ratios do not provide a transparent summary of treatment effects on the survival distribution in the same way as survival probabilities, cumulative incidence, or restricted mean survival time [11].

It is important to distinguish these interpretational challenges from issues of identifiability or estimation. Marginal hazard ratios can be defined and consistently estimated using singly robust approaches, such as unadjusted Cox regression in randomized trials or inverse probability weighted Cox models under correct specification of the treatment or censoring mechanism. In these settings, the estimand is a marginal hazard ratio defined through population-level hazards. However, such estimands remain inherently conditional on treatment-affected risk sets and do not correspond to contrasts of individual-level counterfactual outcomes.

Importantly, these limitations have implications for doubly robust estimation [1]. Doubly robust methods rely on combining outcome models and treatment or censoring models to consistently estimate a single, well-defined target estimand, provided that at least one of the models is correctly specified. Because hazard ratios lack a stable marginal causal interpretation and depend on time-specific, treatment-affected risk sets, there is no natural, time-invariant marginal causal estimand for which the usual notion of double robustness applies in a straightforward way. In particular, while marginal hazard ratios can be defined and estimated using singly robust approaches, they do not admit an individual-level counterfactual representation whose population average defines the estimand. In this sense, the concept of double robustness does not naturally extend to hazard ratios, even when they are derived from marginal survival estimates, because the hazard ratio is not itself a marginal functional of the counterfactual survival distribution.

5 Survival Models

6 Software

Package	Dependencies	Last update	Supported Marginal Estimation Methods	Supported Outcome Models	Supported Estimands
survTMLE [github]		19-nov-2023	TMLE	Discrete-Time • Pooled logistic regression • Super learner	• covariate-adjusted marginal cumulative incidence estimates [14]
Concrete [github]		16-oct-2025	TMLE	Continuous-time • Cox PH	• covariate-adjusted marginal cumulative incidence estimates
adjustedCurves [CRAN]		13-jul-2025	Direct adjustment (with or without pseudo-values); IPTW (multiple variants); AIPW (two variants); empirical likelihood; TMLE[1]	Continuous-time • Cox PH • Cause-Specific Cox-regression • Fine & Gray • Random forests for Survival • Non-parametric/KM estimator Discrete-Time • Binomial regression based on time sequence of binary event status variables	• Adjusted survival curves • Adjusted CIFs (competing risks) • Curve contrasts (difference/ratio) • Adjusted RMST/RMTL [15]
ltmle [CRAN]		15-apr-2023	TMLE	Discrete-Time • Binary (or bounded) outcomes defined at prespecified discrete time points	• Treatment-specific, covariate-adjusted, marginal mean outcomes at specific time points

1 TMLE support depends on availability of the *survTMLE* package, which has been removed from CRAN; see package documentation for version-specific details.

				• Survival handled via monotone discrete-time event indicators	• Risk difference at specific time points • Risk ratio at specific time points • Odds ratio at specific time points
gfoRmula [CRAN]		24-mar-2025	Parametric g-formula for longitudinal and time-to-event data	**Discrete-Time** • Discrete-time outcome models via parametric regression (person–period data)	• Estimated risk at specific time points • Risk difference at specific time points • Risk ratio at specific time points
stdReg2 [CRAN]		28-feb-2025	Regression standardization using parametric models (GLM, Cox PH, frailty)		Covariate-adjusted marginal survival curves; standardized RMST and survival contrasts; marginal risk/mean effects

Other packages exist to estimate conditional survival functions, including survSuperLearner [github], survivalSL [CRAN]. These packages focus on predictive modeling of survival given covariates and do not, by themselves, produce covariate-adjusted marginal survival curves or treatment contrasts with a causal interpretation.

7 Simulation Plan

7.1 Aims

The aim of this simulation study is **to compare robustness of different singly and doubly robust methods that are subject to model mis-specification**. Specifically, we will investigate the performance of these methods when a confounding variable is omitted from either the weighting model (for treatment assignment) or the outcome model, or both. This will aid in obtaining a deeper understanding of which methods are most resilient to such common forms of misspecification i.e. when we have missing confounder in weighting and/or outcome model. We will consider the trade-off between added complexity of each method against the gain in efficiency when model misspecification is present (to various degrees).

7.2 DGM

The data generating mechanism to simulate the population for each iteration will use the following parameters:

- **Sample Size (N):** Each simulated dataset will consist of $N = 2500$ observations
- **Covariates:** A set of continuous time-fixed confounders $X = L_1, \dots, L_6, W_1, O_1$ will be generated from a multivariate normal distribution with mean 0, standard deviation 1, and an 8x8 correlation

matrix that specifies none to high levels of correlation to evaluate the impact of collinearity. A quadratic effect will be included for L_1 and L_2.

- **Exposure Model:** The probability of being assigned to a binary treatment, A, will be estimated using the following model:

$$P(A=1) = \frac{\exp\left(-2 + 0.5L_1^2 + 0.4L_2^2 + (-0.6)L_3 + 0.4L_4 + 0.5L_5 + 0.3L_6 + W_1\right)}{1 + \exp\left(-2 + 0.5L_1^2 + 0.4L_2^2 + (-0.6)L_3 + 0.4L_4 + 0.5L_5 + 0.3L_6 + W_1\right)}$$

- **Outcome Model for Event Times:** The event of interest will be time-to-death. The event times, T_D, are simulated using the uniform inversion method as described by Bender et. al (2005)[16] for cumulative hazards. For the conditional outcome model, we will assume an underlying Weibull model for the baseline hazards with proportional hazards for confounder effects such that,

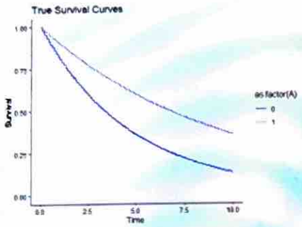
$$h_i(t) = \gamma\lambda(t^{\gamma-1})\exp\left(\beta_0 A + \beta_1 L_1^2 + \beta_2 L_2^2 + \beta_3 L_3 + \beta_4 L_4 + \beta_5 L_5 + \beta_6 L_6 + \beta_O O_1\right)$$

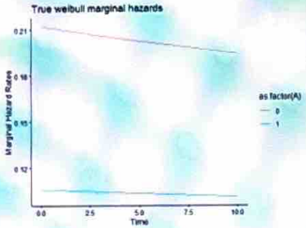
It follows that the baseline hazard function is:

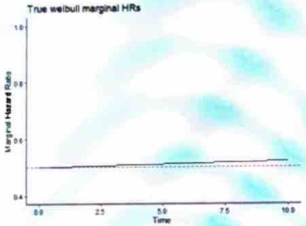
$$h_0(t) = \gamma\lambda(t^{\gamma-1})$$

The scale and shape parameters were selected by trial and error and heterogeneity (impact of confounders) were increased/decreased to achieve the following scenarios for the marginal survival curves i.e. Kaplan-Meier estimates:

- The marginal survival is approximately proportional over follow-up time: $\lambda = 0.5, \gamma = 0.1$

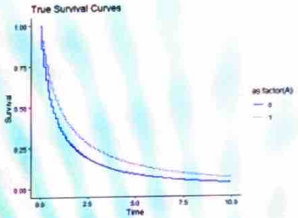


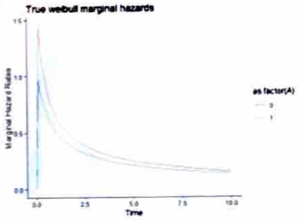


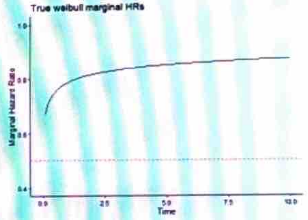


Variable	Coefficient
A	-0.0931
L	-0.0288, 0.0693, -0.0693, -0.0223, 0.1609, 0.0405
L (non-linear)	-0.0051, 0.0405
O	-0.0223
W	0.0405
gamma	1
lambda	0.2

- The curves come closer together over follow-up time i.e. the marginal treatment effect diminishes (non-proportional hazards treatment waning)







Variable	Coefficient
A	-0.6931
L	-0.2877, 0.6931, -0.6931, -0.2231, 1.6094, 0.4055
L (non-linear)	-0.0513, 0.4055
O	-0.2231
W	0.4055
gamma	1.5
lambda	0.9

- There is separation of the curves later in follow-up time i.e. delayed marginal treatment effect (non-proportional hazards delayed effect)

7.4 Methods

- **Methods to be compared:**
  1. Inverse probability of treatment weights – See section 4.1.1
  2. G-computation – See section 4.1.2
  3. Doubly robust standardisation – See section 4.2.1
  4. Augmented inverse probability of treatment weights – See section 4.2.2
  5. Targeted maximum likelihood estimation – See section 4.2.3

- **Outcome Models:** The following models will be used to estimate $h(t \mid A, \boldsymbol{L}, W, O)$:
  - Pooled logistic regression (or a Poisson model): TMLE, AIPW, Doubly Robust Standardisation
  - Cox proportional hazards model: IPTW, G-computation, AIPW
  - Royston-Parmar flexible parametric model with interactions between spline variables and log-time to relax proportional hazards assumption: IPTW, G-computation, DR Standardisation
  - Super learner: AIPW, TMLE

			Estimator		
Outcome Model	IPTW	G-Comp	DR-Stand	AIPW	TMLE
Cox PH (or discrete-time if Cox unavailable)	X	X	X	X	X*
FPM-NPH	X	X	X		
Super Learner	X	X	X	X	X

- **Linear predictor for outcome models:** The analysis models will be mis-specified in various forms in order to evaluate robustness and efficiency of the above methods. By default, both exposure and outcome models will include all confounders $L_1, \dots, L_6$. The following forms of mis-specification will be explored:
  - Missing W_1 in exposure model only. O_1 included in outcome model
  - Missing O_1 in outcome model only. W_1 included in exposure model
  - Mis-specified functional form for L_1, L_2 in outcome model only
  - Mis-specified functional form for L_1, L_2 in exposure model only

7.5 Performance Measures

The following will be reported to evaluate performance of each of the methods:

- **Relative Bias**: The difference between the average estimate across simulations and the true value, expressed as a percentage of the true value.

- **Coverage:** The proportion of simulations for which the 95% confidence interval for the estimate contains the true parameter value.

- **Relative Error:** Calculated as $\left(\left[\frac{model-based\ SE}{empirical\ SE}\right] - 1\right)$, where the empirical standard error is the standard deviation of the estimates across all simulations. This assesses the accuracy of the standard error estimates.

- **Power:** The proportion of simulations in which the null hypothesis of no effect is correctly rejected when a true effect exists.

Each simulation will be replicated 1900 times for a target standard error of 0.5% and for a coverage of 95%.

Covariates/Confounders: $W_1, O_1, L_1, \ldots, L_6$ (multivariate Normal with a specified correlation matrix)
Conditional effect: PH
Model for independent censoring: $T_C \sim Exp(\lambda_{cens})$

Total Scenarios (N)	Degree of Model Misspecification	Marginal effect	Independent Censoring % (λ_{cens})	Dependent Censoring %	Correlation between Confounders	Estimand	Performance measures
72	- Missing W_1 in exposure model only - Missing O_1 in outcome model only	- PH - Delayed - Waning	10 30	0	0 0.25 (Low) 0.75 (High)	- Survival Difference	- Relative Bias - Relative Error - Coverage - Power
	- Mis-specified functional form for L_1, L_2 in outcome model only - Mis-specified functional form for L_1, L_2 in exposure model only						

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Internal CI/ML WG Tutorials: Robust Covariate Adjustment Methodology – Advanced CI/ML Methods

8 Appendix

8.1 Example R code

8.1.1 Confounder-adjusted Survival Curves

```
library(adjustedCurves)
library(ggplot2)
library(survival)

# simulate some data as example
set.seed(31)
sim_dat <- sim_confounded_surv(n=250, max_t=1.2, group_beta=0)
sim_dat$group <- as.factor(sim_dat$group)

# estimate a cox-regression for the outcome
cox_mod <- coxph(Surv(time, event) ~ x1 + x2 + x4 + x5 + group,
                 data=sim_dat, x=TRUE)
```

```
# use it to estimate adjusted survival curves
adjsurv <- adjustedsurv(data=sim_dat,
                        variable="group",
                        ev_time="time",
                        event="event",
                        method="direct",
                        outcome_model=cox_mod,
                        conf_int=TRUE)

# plot with confidence intervals
plot(adjsurv, conf_int=TRUE)
```

8.1.2 Inverse Probability Treatment Weighting (IPTW)

```
# 1) Predicted propensity (no clipping)
g1W <- predict(ps_model, type = "response")    # P(A = 1 | X)
g0W <- 1 - g1W

# 2) Stabilised IPTW
A <- dat$A
pA <- mean(A == 1)                              # marginal P(A = 1)
w_stab <- ifelse(A == 1, pA / g1W, (1 - pA) / g0W)

# 3) Quick weight diagnostics (summary + quantiles + per-group mean)
print(summary(w_stab))
print(quantile(w_stab, probs = c(0, .01, .05, .25, .5, .75, .95, .99, 1)))
print(tapply(w_stab, A, mean))
```

Table 1Use of stabilized weights in IPTW

```
# 1) Truncate stabilised weights at chosen quantiles (e.g., 1% and 99%)
lower_q <- 0.01
upper_q <- 0.99
lower_thr <- as.numeric(quantile(w_stab, probs = lower_q))
upper_thr <- as.numeric(quantile(w_stab, probs = upper_q))
```

```
# Apply truncation (winsorisation)
w_trunc <- pmin(pmax(w_stab, lower_thr), upper_thr)

# 5) Diagnostics for truncated weights
cat("\nTruncated stabilised weights (", lower_q, "/", upper_q, " quantiles):\n", sep = "")
print(summary(w_trunc))
print(quantile(w_trunc, probs = c(0, .01, .05, .25, .5, .75, .95, .99, 1)))
print(tapply(w_trunc, A, mean))
```

Table 2 Use of truncated weigths in IPTW

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